

Automotive Plastic Product Design - Complete Detailed Syllabus

Digital CAD Training | CADPOINT Authorized Partner | Trusted since 2018 | PAN India MNC/OEM/Tier-1 hiring direction

Program Summary

Duration	4 Months / 120 Days
Sessions	80 training + practice roadmap sessions
Core Tools	CATIA V5, UG NX, GSD Surfacing, Draft Analysis, Plastic Trims, GD&T;
Career Focus	Automotive plastic product design, portfolio projects, resume, mock interview, placement support

Sessions 01-10

No.	Session Topic	Category
01	CATIA SURFACING	Surfacing Training
02	CATIA SURFACING	Surfacing Training
03	CATIA SURFACING	Surfacing Training
04	CATIA SURFACING	Surfacing Training
05	CATIA SURFACING	Surfacing Training
06	Automotive Development Process	Theory
07	Benchmarking & Plastic Product Design	Theory
08	Injection Mould & Draft Analysis	Theory + Design
09	Tooling Direction Methods	Theory + Design
10	Class A Surface Analysis	Theory + Design

Sessions 11-20

No.	Session Topic	Category
11	Tree Structure & Close Body	Design
12	Map Pocket Close Body	Design
13	Mastersection Based Close Body	Design
14	Seat Recliner Cover	Design
15	Fuse Box Cover	Design
16	Front Bumper Close Body	Design
17	Front Bumper – Part 2	Design
18	Slider, Lifter & Undercuts	Theory
19	B-Side Features Introduction	Theory
20	Snap Design	Theory + Design

Sessions 21-30

No.	Session Topic	Category
21	Locator Design	Theory + Design
22	Dog House Design	Theory + Design
23	Clip Tower Design	Theory + Design
24	Weld & Screw Boss	Theory + Design
25	Ribs Design	Theory + Design
26	Powercopy	Design
27	RPS & Locating Strategy	Theory + Design
28	Gap & Flush	Design
29	Mastersection Exercise 01	Design Exercise
30	Mastersection EX. 02	Design Exercise

Sessions 31-40

No.	Session Topic	Category
31	Mastersection EX. 03	Design Exercise
32	Mastersection EX. 04	Design Exercise
33	Mastersection EX. 05	Design Exercise
34	Mastersection EX. 06	Design Exercise
35	Mastersection Variable Thickness	Design Exercise
36	Remastering Boolean	Design Exercise
37	Remastering Surfacing	Design Exercise
38	Style Replacement Methods	Design
39	GD and T	Theory
40	Plastic Defects	Theory

Sessions 41-50

No.	Session Topic	Category
41	Plastic Materials	Theory
42	DMU Analysis and Report Creation	Analysis
43	Resume Building and Mock Interview	Career
44	Mock Interview Session 01	Career
45	Mock Interview Session 02	Career
46	Shut off and Graining	Theory
47	2D Drawing creation	Design
48	Features Remastering	Design
49	Mold Flow analysis Report reading	Theory
50	Toolroom mold design review	Theory

Sessions 51-80 - Advanced Practice Projects and Career Readiness

No.	Session Topic	Category
51	Advanced Project Orientation & File Setup	Project Practice
52	Cup Holder CATIA Solution Workflow	CATIA Project
53	B Pillar Upper CATIA Solution Workflow	CATIA Project
54	Seat Recliner Cover NX Solution Workflow	NX Project
55	Armrest Project CATIA Solution Workflow	CATIA Project
56	B Pillar Upper NX Solution Workflow	NX Project
57	Map Pocket Design CATIA Solution Workflow	CATIA Project
58	Map Pocket Design NX Solution Workflow	NX Project
59	Cup Holder NX Solution Workflow	NX Project
60	Fuse Box Cover CATIA Solution Workflow	CATIA Project
61	Mastersection Practice NX Workflow	NX Project
62	Project Tree Structure & Naming Standards	CAD Standards
63	Input Data Study: Class-A, Master Section & Packaging	Design Method
64	Tooling Direction & Draft Strategy for Practice Projects	DFM
65	Close Body Quality Check & Surface Continuity	Quality Check
66	B-Side Feature Placement Review	Design Review
67	Snap, Locator, Dog House & Rib Validation	Design Review
68	Assembly Clearance, Gap & Flush Check	DMU
69	CATIA to NX Method Comparison	Tool Comparison
70	Project Error Debugging & Rework Methods	CAD Debugging
71	Drawing, Screenshots & Report Preparation	Documentation
72	DFM Checklist for Plastic Trim Projects	DFM
73	Mould Feasibility Review for Project Parts	Mould Review
74	Portfolio Sheet Creation for Each Project	Portfolio
75	Interview Explanation: Project 01 to 03	Interview Prep
76	Interview Explanation: Project 04 to 06	Interview Prep
77	Interview Explanation: Project 07 to 10	Interview Prep
78	Mock Technical Review With Project Questions	Mock Interview
79	Resume Project Description Writing	Career
80	Final Portfolio Review & Job Readiness Checklist	Career

Solution Project Practice Library

These projects are included as solution/practice references for CATIA and NX based project learning.

No.	Project Folder / Solution Name
01	PROJECT 01 - CUP HOLDER CATIA SOLUTION
02	PROJECT 02 - B PILLAR UPPER CATIA SOLUTION
03	PROJECT 03 - SEAT RECLINER COVER NX SOLUTION
04	PROJECT 04 - ARMREST PROJECT - CATIA SOLUTION
05	PROJECT 05 - B PILLAR UPPER - NX SOLUTION
06	PROJECT 06 - MAP POCKET DESIGN CATIA SOLUTION
07	PROJECT 07 - MAP POCKET DESIGN NX SOLUTION
08	PROJECT 08 - CUP HOLDER NX SOLUTION
09	PROJECT 09 - FUSE BOX COVER - CATIA SOLUTION
10	PROJECT 10 - Mastersection Practice NX

Placement and Trust Direction

The program is designed for mechanical freshers, production/quality engineers, CAD beginners and career switchers who want practical automotive product design exposure. Digital CAD Training communicates CADPOINT authorization, 7+ years trust since 2018, PAN India MNC/OEM/Tier-1 hiring direction, portfolio development, resume support and mock interview preparation.